



SIR System-30

Manual

Geophysical Survey Systems, Inc.
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This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Warning: Changes or modifications to this unit not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment or residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the introduction manual, may cause harmful interference to radio communications. However, there is not guarantee that interference will not occur in a particular installation.

Shielded cables must be used with this unit to ensure compliance with the Class A FCC limits.

Canadian Emissions Requirements

This Class A digital apparatus complies with Canadian ICES-003.

Cet appareil numérique de la classe A est conforme a la norme NMB-003 du Canada.

Notice

Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

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Chapter 1: Introduction

Thank you for purchasing a SIR-30 GPR control unit from GSSI. The SIR-30 represents the current state of the art in very high speed multi-antenna GPR. The SIR-30 is designed to work with any of GSSI's current and past line of GPR antennas allowing you to more easily add the SIR-30 into your existing GPR system configuration. The SIR-30 also offers unparalleled integration options, so it is easy to add high speed GPR to a multi-method data collection vehicle.

Please take a moment to unpack your order and ensure that everything is present. Contact GSSI or your local GSSI representative immediately if you notice anything missing or damaged. GSSI can be reached at (603)893-1109, 8:30 AM – 5 PM US Eastern Time, Monday-Friday. You can also email sales@geophysical.com.

GSSI suggests that you read completely through this manual before using your new SIR-30.

Unpacking the Box

Your new SIR-30 comes in a Pelican Model 1650 transit case with custom-cut protective foam. The transit case is designed to protect your SIR-30 from the elements and from damage. GSSI recommends keeping your SIR-30 in its protective case when not in use.

The basic SIR-30 system comes with:

- **SIR-30 controller:** This is the blue box with all of the connectors and “SIR-30” printed on it.
- **AC Power supply:** This is for powering the SIR-30.
- **Manual:** The document which you are reading right now.
- **Documentation CD:** This CD contains an electronic version of this manual as well as all of GSSI's other manuals.
- **SIR-30 Maintenance Tool.** This is a USB Key containing tools to maintain the system and install updates to your SIR-30 user interface.
- **Transit Case:** The Pelican 1650 case that all of the above came in.

You may also want to consider some accessories to your SIR-30. These are available from GSSI for a separate purchase. Optional accessories include:

- A Panasonic Toughbook laptop computer pre-configured to run the SIR-30.
- Rugged Ethernet cross cable (3 or 7 meter lengths) for connecting the SIR-30 to a laptop computer.
- DC power cable with battery clips. This will allow you to power the SIR-30 from a 12v battery.
- Rugged HDMI cable (2 meters) for connecting the SIR-30 to a monitor.
- Universal mounting kit for securing the SIR-30 when used in a vehicle.
- RADAN 7 data processing and interpretation software.

Key New Features of the SIR-30

GSSI would like to call your attention to some key new features of the SIR-30.

Speed: The SIR-30 can transmit at a maximum pulse repetition frequency (PRF) of 733 KHz. That means the SIR-30 is capable of collecting nearly three-quarters of a million samples each second. The PRF is not degraded by using multiple channels at the same time. This means that you will be able to collect more data faster than ever before. Please note that not all antennas are capable of operating at that speed. Be sure to consult your antenna documentation for the correct antenna PRF. Setting the SIR-30 PRF higher than the maximum antenna PRF will result in poor data quality. Also note that your local regulatory environment may limit the SIR-30's maximum PRF.

Data resolution: The SIR-30 incorporates 32-bit sampling. Other commercially available GPR systems sample at 16-bit. This means that the SIR-30 has a vastly superior dynamic range which means that you will see weaker signals with SIR-30. The SIR-30 is also applying filters and gain during data collection *for display purposes only*. Data are stored completely raw which means that it is now impossible to “clip” data by incorrectly over-gaining the signal. The filters and gain that are stored with the data file can be automatically applied in RADAN 7 if you desire.

SIR-30 Physical Characteristics and Connections

Front Panel Connectors

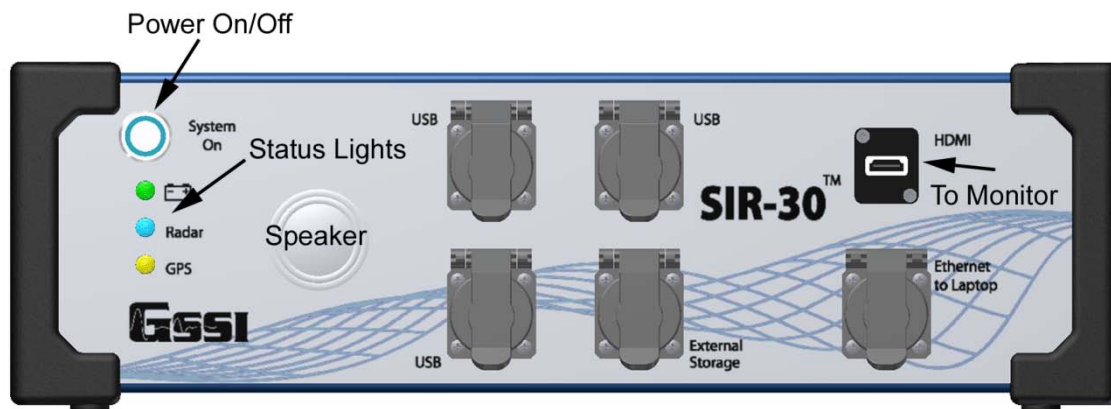


Figure 1: SIR-30, Front

The front panel of the SIR-30 has the more frequently accessed connectors as well as the status lights (Figure 1).

Power On/Off: This button will turn the SIR-30 on and off. To turn on the SIR-30, push this button in so that it stays depressed. The SIR-30 will begin its boot up sequence. *If you are running from a separate laptop, for approximately 10 seconds there will be no indication on the SIR-30 that something is happening.* Please be patient. After approximately 10 seconds, the blue ring around the button will illuminate and you will hear the SIR-30 fans. The SIR-30 is ready when you hear a short tone. Push this button again to turn off the SIR-30. *For approximately 30 seconds there will be no indication that something is happening.* Please wait for the blue ring light around the power button to go out before unplugging the SIR-30.

Status Lights: The green LED next to the battery icon will stay lit when the SIR-30 is connected to a power source. This light will be on even if the SIR-30 is turned off. The blue LED next to the word “Radar” will be lit when the internal GPR electronics are powered up. The amber LED next to “GPS” will be lit and flash if the SIR-30 detects incoming GPS data.

USB ports (3): These are for connecting various USB peripherals such as a mouse and keyboard.

External Storage: This is an Ethernet port is for connecting external storage devices such as an Ethernet hard drive. This may be a USB Port on some earlier units.

HDMI: This port is for connecting an external monitor. Note that you will need to power that monitor separately.

Ethernet to Laptop: This port is for connecting the SIR-30 to a computer via an Ethernet link. Note that you will need to power that computer separately.

Rear Panel Connectors



Figure 2: SIR-30, Rear

Accessory: This port is for connecting custom accessories to the SIR-30. Contact your GSSI sales representative for details.

Power In: Connect the SIR-30 power supply to this port.

Sync: This is for connecting multiple SIR-30's in order to create an 8-channel array.

Survey Wheel: This is for connecting any GSSI 4-pin survey wheel or DMI connector. This connector may be repeated on your antenna.

Marker: This is for connecting a GSSI 2-pin marker switch. This connector may be repeated on your antenna.

GPS: This port is a 9-pin serial (RS232) that you can plug your GPS receiver into. Be sure to configure the output on your GPS according to the specifications in the section of this manual on GPS setup.

Channel 1-4: These ports are for connecting the antenna control cables. Try to keep the caps on these when not in use in order to keep them free from dirt. There is a LED above each connector which will illuminate when an antenna is connected and powered up. Be sure to only hand-tighten these as over-tightening may result in damage.

Note: Do not plug in or unplug an antenna while the SIR-30 is running. Doing so may result in damage to the SIR-30.

SIR-30 Hardware Setup

The SIR-30 can be run directly or through a laptop.

Running the SIR-30 Directly

The SIR-30 is a Windows PC and can be run without a separate laptop. You will need a HDMI monitor, USB keyboard and USB mouse (not supplied).

With the SIR-30 powered OFF:

- 1** Plug the HDMI cable from your monitor into the HDMI port on the front of the SIR-30 and plug your monitor's power cable into a power supply. Turn on the monitor.
- 2** Plug the USB cables from your keyboard and mouse into the USB ports on the front of the SIR-30. It doesn't matter which USB port is used for the keyboard and mouse.
- 3** Plug the control cables from your antennas into your desired channels on the back of the SIR-30.
- 4** Ensure that your SIR-30 is plugged into a power supply and that the green LED on the front is illuminated.
- 5** Push the System On button to turn the SIR-30 on. You will see the boot up sequence on the screen. The SIR-30 is fully booted when you see the Windows desktop.

Running the SIR-30 from a Laptop PC.

The SIR-30 can be controlled from a separate PC running the SIR-30 user interface. The look and feel of the software will be the same as running directly. You will need to power the computer separately.

Note: Minimum requirements:

- Intel Core i3 or better
 - 3GB System Memory
 - Windows 7 or Vista (except Home or basic versions)
 - nVidia or Intel Video chipsets
- 1** Load the SIR-30 User Interface installation disk into your PC and follow the onscreen instructions to install the necessary software onto your laptop. You can skip this step if you purchased a companion Panasonic Toughbook from GSSI.
 - 2** Connect your Laptop to the SIR-30 with an Ethernet cable. Maximum allowable cable length is 50 m (150 ft).
 - 3** Turn on the Laptop and wait for it to fully boot.
 - 4** Push the System On button to turn the SIR-30 on. Remember to wait 10 seconds after pushing the button for the SIR-30 to come on. You will hear a tone once the SIR-30 is completely booted up.

Chapter 2: User Interface Descriptions

This section will familiarize you with the various menus on the SIR-30. GSSI recommends that you read through this section so that you gain an understanding of the menu structure and logic.

System Settings Screen

After clicking on your desired units, you will see the System Settings screen shown in Figure 3. From here you can tell the SIR-30 about the geometry of your collection setup, location of the GPS receiver, and also your desired scan density and mark spacing.

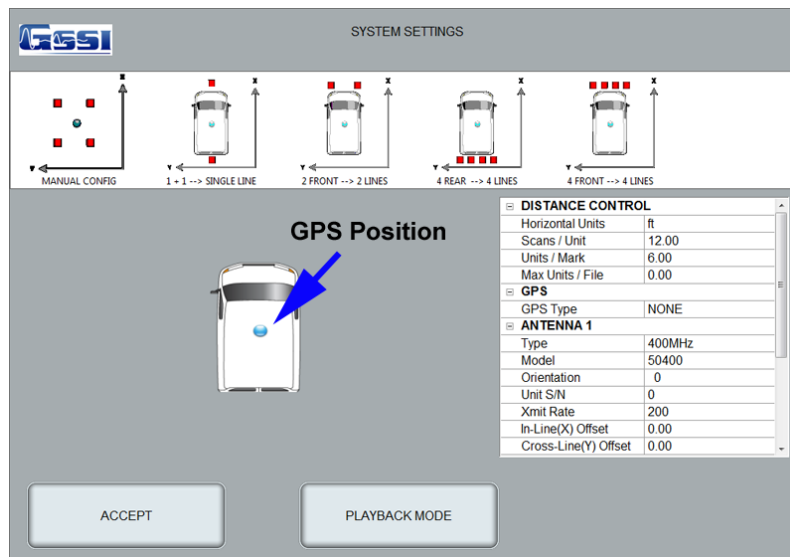


Figure 3: Initial System Settings

Distance Control: The scan density will be in whichever units are selected in the Horizontal Units option. In Figure 3, the option shown is 24 scans/ft. This option should be set dense enough to effectively image what you are looking for yet not so dense as to create too much data. Common scan density settings for different applications may be found in Appendix A.

- The Unit/Mark setting determines how often the SIR-30 will write a distance mark in the data. Distance marks are used as quick visual indicators of distance.
- Max Units/File. This is the total expected length of the individual data file. When you start a data file, the SIR-30 reserves a section in the memory to store that file. This value is used to make sure it reserves enough room. This number doesn't need to be exact. If you leave this at 0, the SIR-30 will default to a large value equivalent to 5 GB per channel.

GPS: If you are using a GPS, select Custom and enter the location of the GPS receiver with respect to the antenna connected to Channel 1. If it is mounted to the antenna, both the X and Y values will be 0. If the GPS is behind the antenna, the X value will be the distance behind the antenna, and the number will be *negative*. The Y value determines how far to the right or left the GPS is relative to the antenna connected to Channel 1. If you are standing behind the antenna, facing the direction of your survey profile, this number is *positive* if the GPS is to the *left* of the GPR antenna and *negative* if the GPS is to the *right* of the GPR antenna.

Antenna Entries. Here is where you will enter your antenna selections. The “Antenna 1-4” options correspond to the Channel 1-4 inputs on the back of the SIR-30. For each antenna, choose the Type (Frequency), antenna model (some will only have one available), orientation (angle relative to your profile direction), serial number and antenna location relative to the antenna connected to Channel 1. The X and Y values will be 0,0 for Antenna 1.

- If you are using one antenna, these will be 0,0.
- The serial number is important information if you are using a Model 4105NR 2 GHz Horn with noise reduction technology because the SIR-30 will use the provided serial number to select the correct preloaded custom filter file.
- Clicking **Accept** will cause the antenna(s) to initialize and display the Control Screen shown in Figure 4.
- If you selected an antenna model, the SIR-30 will automatically load the maximum possible transmit rate (abbreviated Xmit Rate). If you would like to set this manually, you must select “Custom” from the Type pulldown menu. Although you can specify different transmit rates for each channel, the SIR-30 will use a common transmit rate across all channels which will be equal to the slowest channel. See Appendix B for a list of maximum antenna transmit rates.

Control Screen

Figure 4 shows the Control Screen with a simple, one antenna case. Clicking the Channel#/Antenna Model button will activate that channel.

- Once activated, you will see a linescan view extend across the middle of the screen and any changes you make in the setup controls will apply to the activated channel.
- If you have more than one channel, you should take care to activate the particular channel before making any changes to that channel in the setup controls.

Main Menu

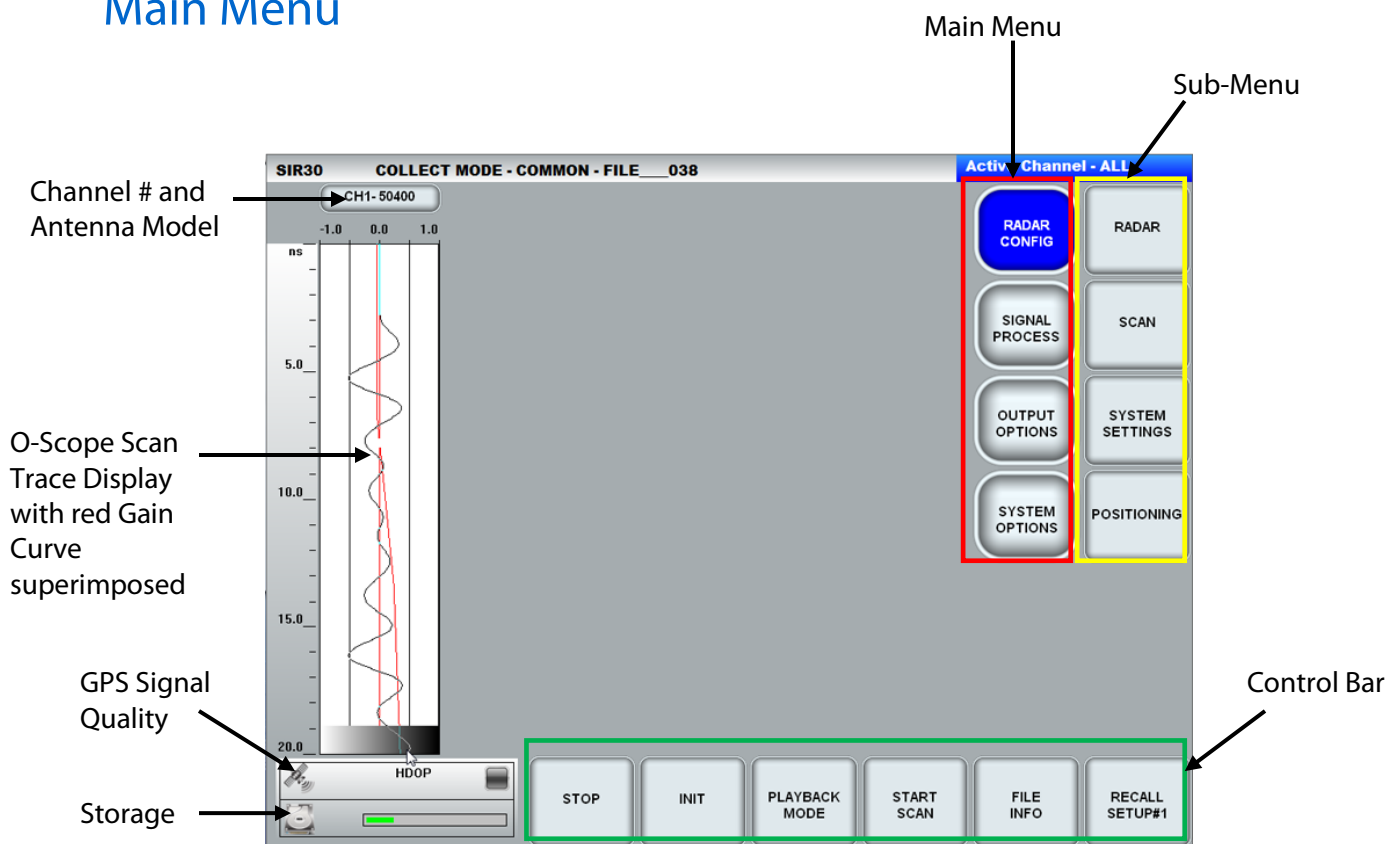


Figure 4: Control Screen

For this section, we will refer to the left hand column of the Setup Controls as the Main Menu and the right hand column as the Sub-Menu. The SIR-30 user interface is designed so that you will move through the control settings in a linear manner from top to bottom. When you click one of the Main Menu buttons, it will turn solid blue and the options in the Sub-Menu will change.

There are four buttons in the Main Menu:

- RADAR CONFIG (page 7)
- SIGNAL PROCESS (page 10)
- OUTPUT OPTIONS (page 12)
- SYSTEM OPTIONS (page 13)

RADAR CONFIG

Clicking this Main Menu button will display four additional buttons in the Sub-Menu:

- RADAR
- SCAN
- SYSTEM SETTINGS
- MAINTENANCE

RADAR CONFIG

RUN MODE: This option will allow you to select Time, Distance OR Point. Use Time if you are collecting data without a survey wheel or distance measuring instrument (DMI). Selecting Time will cause the SIR-30 to collect data at the scan rate listed in the Scans/Sec option in the RADAR Config/RADAR sub-menu. Your scan density over the ground is thus a function of how fast you move your antenna.

Selecting Distance will cause the SIR-30 to collect data at the scan density that was entered in the Scans/Unit option in the Systems Settings screen (Figure 3).

Select Point if you are using large, low frequency antennas, are working on uneven terrain, or performing specialized techniques such as CMP collection. In Point mode you will be collecting on scan at a time. Selecting Point will cause the Static Stacking option to appear in the RADAR Config menu.

SCANS/SEC: This value will be automatically set and is a function of the transmit rate and the Samples/Scan. This value should be set as high as possible for collection with Run Mode: Distance. You may wish to set this value lower while collecting in Run Mode: Time in order to avoid collecting unnecessarily large files.

SAMPLES/SCAN: Each scan is digitized with a discrete number of samples evenly spaced along the time range. Setting this number too low will result in a problem called “aliasing.” Aliasing means digitizing the signal at a lower frequency than it really is due to an inadequate number of samples. This value should always be equal to or greater than $(10(\text{time range})) / (\text{antenna's pulse width})$. See Appendix B for a listing of antenna pulse widths. Setting this number too high will result in unnecessarily large files.

For example: A 400 MHz antenna has a pulse width of 2.5 nS. Using the formula and collecting at a range of 100 nS would mean that a sample density of 400 per scan or greater is required. Since the list of sample densities runs from 256, 512, 1024, etc., you would then set the SIR-30 to 512.

STATIC STACKING: This option is only visible if you have selected Run Mode: Point. This will allow the SIR-30 to collect *n* scans at a given location, average them together and output the result. This may result in a higher signal to noise ratio and deeper penetration. If you are collecting in Point mode, GSSI recommends setting this as high as practical.

SCAN

TIME RANGE: This value is the two-way travel time (TWTT) measured in nanoseconds that the SIR-30 will use for the selected channel. A larger value will allow the SIR-30 more time to record reflections that may come from farther away from the antenna. The actual scanning depth is a function of time range and the dielectric constant of the material you are scanning through.

DIELECTRIC CONSTANT: This value is used for a simple time to depth calculation that will update the vertical scale of the O-Scope scan trace. While it has no effect on the actual collection depth, selecting a value which accurately reflects the dielectric constant of the material you are scanning through will result in reasonably accurate depth values showing on the vertical scale. If you choose to use the Migration filter in Signal Process > Advanced Filters, it will use this value to perform the Migration process. This value can be modified in post-process software. Possible values range from 1 (air) to 81 (water). See Appendix C for a listing of common materials and their dielectric values.

POSITION MODE: Possible choices are Auto and Manual. This will allow you to set Time-Zero automatically or manually. Time-Zero is the beginning of the time range.

POSITION OFFSET (nS): This value is the location of Time Zero with respect to the hardware zero of the antenna. Each antenna model is different, so this value is best set by eye. Figure 5 shows two 1.6 GHz, Model 5100 antennas. The antenna signal on the left is correctly positioned with the first positive peak of the direct wave near the top of the o-scope window. It is important to leave a short section of zero-amplitude signal above the direct wave.

SURFACE (%): The SIR-30 will crop off the top part of the visible scan trace for display purposes only. The value here is the percent of the time range hidden. For example, a value of 10% and a time range of 50 nS will crop the top 5 nS of the visible trace. You can only set the Surface % to Zero if you are running in the Manual position mode. The automatic setting is a 5% (recommended).

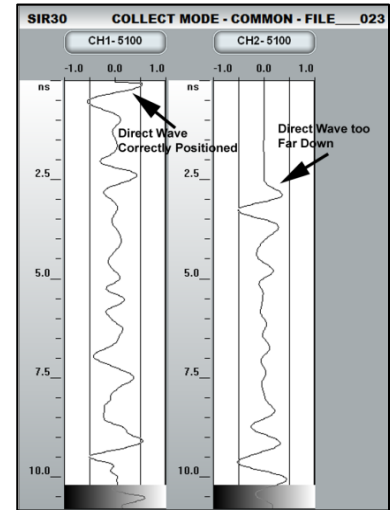


Figure 5: Position Offset

SYSTEM SETTINGS

Clicking this button will return you to the initial System Settings screen shown in Figure 3.

MAINTENANCE

CALIBRATE SW: This will allow you to calibrate your cart, survey wheel or DMI. GSSI recommends calibrating your device as often as practical. To calibrate the wheel, lay out a distance with a tape measure or some other accurate measuring device. Enter that distance in the appropriate box. Position the antenna at zero and click start. Move the antenna to the end of your measured distance and click stop. You will see the Tick Count change while you are moving. After clicking stop, you will see the new calibration value displayed. This represents number of counted ticks inside the survey wheel encoder per unit distance of travel. You should repeat this process to make sure you get a consistent number. All GSSI supplied carts, survey wheels and DMI's operate in Quadrature mode. If you are using a non-GSSI supplied device, contact your device's manufacturer to find out if it operates in Quadrature or Pulse mode. Note that a quadrature device will count up and down while a pulse device will only count up.

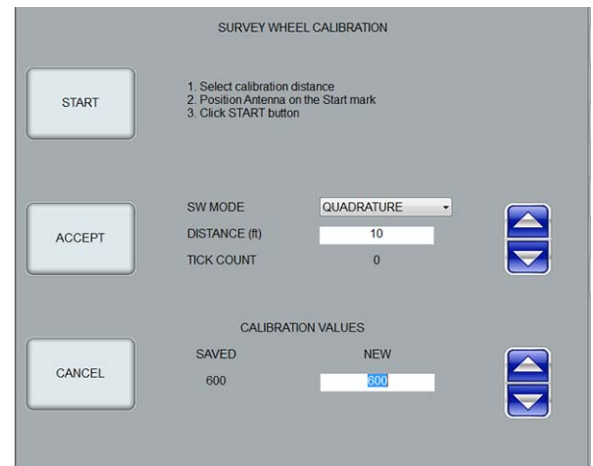


Figure 6: Survey Wheel Calibration

CONFIGURE GPS: This button allows you to enter configuration settings for your GPS. Please see the section of this manual pertaining to GPS configuration for more details.

CUSTOM MARKS: The SIR-30 is able to record and display up to ten different types of User Marks which may be used to denote different conditions or events noticed during survey. You can click this button and enter in the name of the mark type corresponding to one of the keyboard function keys. This information will be stored in the data file's marker table.

DSP CONFIG: Clicking this will display various internal system addresses. GSSI does not recommend changing any of these. You can also exit the SIR-30 program from here.

SIGNAL PROCESS

Clicking this Main Menu button will display four additional buttons in the Sub-Menu:

- GAIN
- IIR FILTERS
- FIR FILTERS
- ADVANCED FILTERS

The SIR-30 collects and stores raw data. Any settings input in this menu will be applied for display purposes and stored as a file with the same root name of the GPR data file but with the extension .DZW. For example, if you are collecting FILE____019.DZT, the settings file will be FILE____019.DZW. When you open the DZT file in RADAN 7, you can choose to apply those settings to the data.

NOTE: Make sure that you activate a channel by clicking on the Channel # button above the appropriate o-scope trace. GSSI recommends that you make changes to only one channel at a time.

GAIN

Gain is artificial signal amplification used to counteract the natural effects of attenuation in the ground. Gain can be applied at 1-8 evenly spaced locations along the time range. If one gain point is used, it will be located at 0 nS. Possible values range from -42 to 126. Because the SIR-30 collects 32-bit raw data, it is not possible to clip your data by over gaining. Figure 7 shows a properly gained signal. The blue arrows show the width of 50% of the *visible* dynamic range. Note that the Gain Mode will need to be set to Manual if you wish to change the values. GSSI recommends that each gain point be greater than or equal to the previous point. The red line superimposed on the scan trace is a representation of the gain curve.

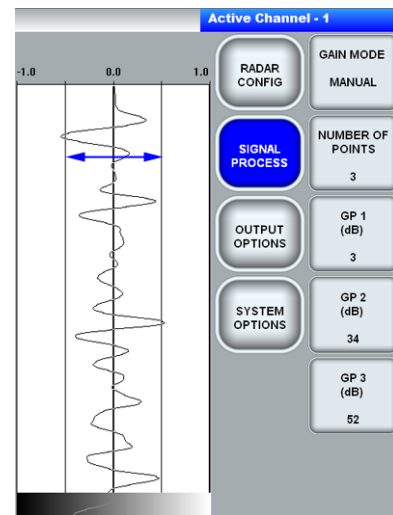


Figure 7: Gain Settings

IIR FILTERS:

IIR stands for Infinite Impulse Response. These are a class of filters used to remove noise and improve the signal quality.

LOW PASS IIR: The value here is expressed in frequency (MHz). A low pass filter will pass everything lower than a given value, so this number should be high. Common values are 2 or 3 times the antenna's center frequency (800 MHz for a 400 MHz antenna).

HIGH PASS IIR: The value here is expressed in frequency (MHz). A high pass filter will pass everything higher than a given value, so this number should be low. Common values are 1/4 or 1/3 the antenna's center frequency (100 MHz for a 400 MHz antenna).

STACKING: This is a horizontal filter where the value is expressed in scans. This will smooth the data and dampen random noise. This value should be set very low. Common values are 3-7 scans.

BACKGROUND REMOVAL IIR: This is a horizontal filter where the value is expressed in scans. This filter will remove flat, horizontal bands in the data which may be associated with ringing noise. The filter is a moving window which will average n scans and then remove that average scan from all scans. This filter value should be set very high otherwise you may remove actual data. Common values may be 300+.

FIR FILTERS

FIR stands for Finite Impulse Response. These are a class of filters used to remove noise and improve the signal quality. They operate in the same manner as the IIR filters, but they often produce a better result and do not shift the signal in phase or amplitude. GSSI generally recommends the use of FIR filters as opposed to IIR filters.

LOW PASS FIR: The value here is expressed in frequency (MHz). A low pass filter will pass everything lower than a given value, so this number should be high. Common values are 2 or 3 times the antenna's center frequency (800 MHz for a 400 MHz antenna).

HIGH PASS FIR: The value here is expressed in frequency (MHz). A high pass filter will pass everything higher than a given value, so this number should be low. Common values are 1/4 or 1/3 the antenna's center frequency (100 MHz for a 400 MHz antenna).

CUSTOM FILTER: This is a special filter used on any GSSI antenna with the NR (noise reducing) designation in its model number. Some models of GSSI Horn antennas use this filter. This filter supersedes all of the other filters (IIR and FIR) and those other types should not be used when the custom filter is used.

Note: The custom filter is supplied by GSSI and must be pre-loaded onto the SIR-30. To apply the filter, make sure that you have specified a serial number for the antenna in the System Settings screen (Figure 3).

STACKING (FIR): This is a horizontal filter where the value is expressed in scans. This will smooth the data and dampen random noise. This value should be set very low. Common values are 3-7 scans.

BACKGROUND REMOVAL FIR: This is a horizontal filter where the value is expressed in scans. This filter will remove flat, horizontal bands in the data which may be associated with ringing noise. The filter is a moving window which will average n scans and then remove that average scan from all scans. This filter value should be set very high otherwise you may remove actual data. Common values may be 300+.

ADVANCED FILTERS

These are a class of specialized filters that are less commonly used during data collection. You should still apply an FIR or IIR Filter if you intend to use these.

Note: Do not use any of the Advanced Filters if you have selected a custom filter in the FIR Filters sub-menu.

SIGNAL FLOOR: This filter will analyze the data's signal to noise ratio and place a solid line at the time-depth where the data becomes marginal. This is to assist you in recognizing the maximum extent of your usable data. This filter is saved in the DZW file and can be reviewed in RADAN.

MIGRATION: This filter will downplay diagonal banding and focus the hyperbolas into peaks. This may improve the visibility of point targets such as rebar or pipes. You will need to specify a Dielectric Constant value in RADAR Config > Scan. A correct value will collapse the hyperbolas into small dots. Migration works best on homogenous materials such as concrete.

ADAPTIVE BKGREM: This filter is designed to be used with the Migration process turned on. It will remove everything except high amplitude reflections from point targets such as rebar or pipes.

POSITION TRACKING: This filter is designed to compensate data position for any temperature related drift of the Time Zero. Surface position signature (defined as the data at SURFACE distance from the beginning of the scan) is being corrected in real time. The scans are being shifted up or down to keep surface position always at fixed sample.

OUTPUT OPTIONS

DISPLAY

This sub-menu contains six buttons which control the way the data will look on the screen. None of these functions are stored in the DZW file and are only for viewing on the SIR-30.

- COLOR TABLE
- COLOR TRANSFORM
- DISPLAY GAIN
- CHANNEL SPLIT
- TICKS PER UNIT
- TICKS PER MAJOR TICK



Figure 8: Color Tables

COLOR TABLE: Clicking this button will bring up the list of available color tables (Figure 8). Click on one of the color bars to choose your desired color table.

COLOR TRANSFORM: This option will allow you to choose the distribution of the color table's shades across different segments of the amplitude scale. For example, if you want to accentuate weaker, low amplitude reflections, you should choose a color transform that shows more variation in the center of the color bar. This corresponds to zero amplitude so that a weak reflection will produce a noticeable shade change.

DISPLAY GAIN: Display gain is screen contrast. The values are expressed in dB. You can choose to increase or decrease the gain. All portions of the screen will change by the same amount so this may result in some sections being over or under gained relative to other sections. Positive values will increase the contrast, negative will decrease.

CHANNEL SPLIT: There are two options here: Equal and Scale. Selecting equal will give the same amount of display space to each channel regardless of time range. This means that if one channel is collecting 12 nS and the other is collecting 50 nS, the vertical scales will be altered to take up the same amount of screen room. If you select Scale, the screen will appear as in Figure 9.

TICKS PER UNIT: This will allow you to space the system marks (Major Ticks) at different intervals.

TICKS PER MAJOR TICK: This option allows you to space the minor ticks at varying intervals.

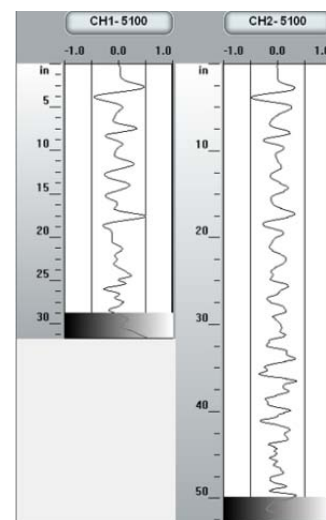


Figure 9: Scaled Channels

MANAGE FILES

DATA PATH: Here you will see a listing of previously created folders with their size and modification date. The default path is “Common.” If Common is selected, data files will have the root name “FILE_###” where “###” is the sequential numerical designation of the file. You can create a new folder by clicking “New Data Path” and typing in a name. New data files will be stored here and this new name will replace “FILE” in your data file’s root name. To collect additional data files to a previously created data path, just select it and click “Accept.”

COPY/MOVE TO LAPTOP: You will see these buttons if you are running the SIR-30 UI on a laptop. Data are stored on the SIR-30 internal hard drive and will need to be copied or moved to that laptop. Before attempting to copy or move data, you must make sure that there is a folder on your laptop with the name “radardata”. Make sure that this folder is on the computer’s “Local Disk (C:)”. This folder cannot be in “My Documents” or on the Desktop. This is the destination folder for the data. If you select “Move” the SIR-30 will delete the data file from its memory after it has been transferred to the laptop. GSSI recommends using the “Copy” command, ensuring that the file is on the laptop, and then manually deleting it from SIR-30.

COPY/MOVE TO DRIVE E/F: Drive E is a specially designated external storage drive for the SIR-30. It can be a USB key plugged into the laptop or an external hard drive plugged into the SIR-30 External Storage port.

DELETE FILES: Select the files you wish to delete and then click Accept. The SIR-30 will not ask “Are you sure?” so be sure. Note that this will only delete files in the currently selected Path.

SYSTEM UNITS:

This menu has 3 options to allow you to change the Depth/Distance units and to display the Depth Scale (vertical scale) as either Time or Depth. If you specify depth, the SIR-30 will use the Dielectric Constant that you specified in the RADAR Config>SCAN menu to do the time to depth calculation.

SYSTEM OPTIONS

This menu has various default settings and version information.

SET TIME: This will open the Window OS date and time set window. Your data files will be tagged with the date and time you specify here.

LANGUAGE: The SIR-30 may be displayed in different languages. GSSI will add different language files as the SIR-30 user interface is translated. Contact GSSI for details.

VERSIONS: Check here to see if your SIR-30 software and firmware are up to date.

CONFIG SYSTEM: This window contains various addresses and internal configuration options for the SIR-30. GSSI does not recommend modifying these without a GSSI approved service technician. You can also exit the SIR-30 application from this window.

SAVE/RECALL SETUP: If you are going to be using common settings for a long project or multiple projects, you can save them by selecting one of the sixteen saved setup slots. You could also click “New Setup” and type a unique name for the new setup. Recall setup will allow you to choose a previously saved setup.

CONTROL BAR (SETUP)

You will see the row of buttons shown in Figure 10 while the SIR-30 is in Setup mode.



Figure 10: Control Bar during Setup

PREVIEW SCAN/STOP: The SIR-30 will collect, but not save, data while in setup mode in order to give you the opportunity to see the effect that any change to the settings may have.

INIT: This stands for “Initialize.” Clicking this will cause the channel(s) to re-initialize with the current settings.

PLAYBACK/COLLECT MODE: Click this to toggle to Playback mode and open a file chooser window.

START SCAN: Clicking this button will cause the SIR-30 to begin collecting data.

- **PLAYBACK FILE(S):** While in Playback mode, the “Start Scan” button will say Playback File(s). Clicking this will cause the SIR-30 to scroll the whole data file and then stop at the end.

FILE INFO: Clicking this button will display the file header which contains antenna description and setup information as well as any notes that you might want to type it. You can also give the data file a unique name by typing it in the File Name window. If you want to start your file at a non-zero origin point, type in the position in the Start Pos box. In addition, if you want to have the SIR-30 count up from that point (default) or down, make the selection in the Direction menu. Having the SIR-30 count down would be especially appropriate for Roadscan applications where you would like the data to match a mile/km marker.

RECALL SETUP#1: This is a shortcut to recall a single setup. If you have a common list of settings that you use repeatedly, you can name that setup “SETUP#1” and easily recall it with this button. Goto SYSTEM OPTIONS>SAVE SETUP and name it SETUP#1. If there is already one there, select it and overwrite it.

CONTROL BAR (RUN)

After you click “Start Scan” in Collect mode or “Playback File(s)” in Playback mode, the Control Bar will change to look like Figure 11.



Figure 11: Control Bar (Run)

PAUSE/RESUME SCAN: Clicking this will pause your data collection or scrolling (in Playback). This is especially useful if you are collecting data in Time mode.

NEXT FILE: Use this button if you wish to stop your current file and collect another one with the same settings. After you click this button, a box will pop up and you will be asked to Save or Discard your data file.

ACTIVE CHANNEL: This button will toggle the active display through the channels so that you can see them individually or all at the same time while collecting.

STOP SCAN: Clicking this button will end data collection, bring up a save menu, and return you to the setup screen.

GAIN (up/down): This is display gain. Any changes you make will not affect the saved data. Display gain will also only be applied to newly collected data. It will not be applied to data already on the screen.

Chapter 3: Data Collection

Setting Up the SIR-30 for Data Collection (General).

When you turn on the SIR-30 please remember that there is a an up to 10 second lag between pushing the “Power On” button on the SIR-30 and a indicator on the SIR-30 that the system is powering up.

This section will cover general setup information. If you are using GSSI’s air-coupled Horn antennas for asphalt thickness measurement, please familiarize yourself with this section and also the section following this one.

This section assumes you are collecting data in Distance mode with a survey wheel or DMI.

Once the system is booted up, and if you are using a separate laptop, double-click on the SIR-30 UI icon for either “English” or “Metric” units. You will not be able to change measurement systems while in the SIR-30 collection program so be sure to choose the proper system.

If you are running the SIR-30 direct, select “English” or “Metric” from the startup screen.

Step 1: Physical System Settings

- 1 After clicking on your desired units, you will see the screen shown in Figure 12. From here you can tell the SIR-30 about the geometry of your collection setup, location of the GPS receiver, and also your desired scan density and mark spacing.

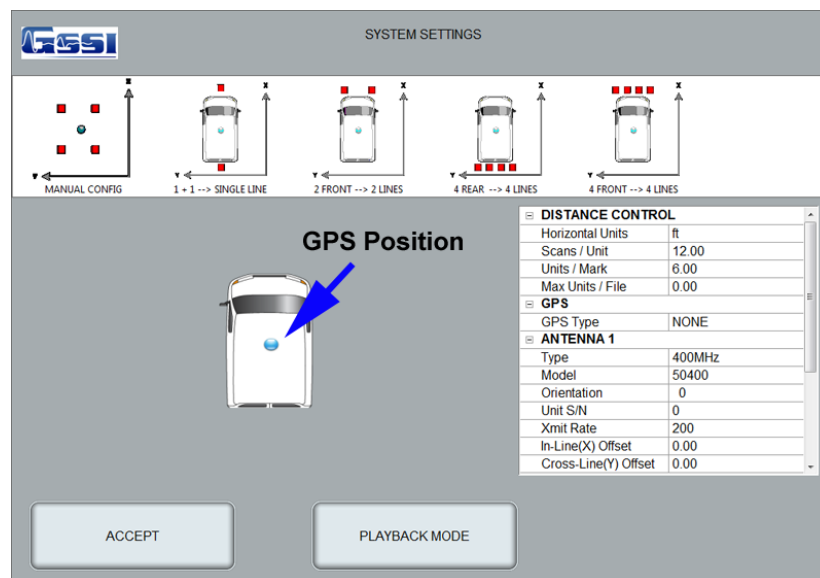


Figure 12: Initial System Settings

- 2 **Distance Control:** Choose your scan density and mark spacing. The scan density will be in whichever units are selected in the Horizontal Units option. In Figure 12, the option shown is 24 scans/ft. Common scan density settings for different applications may be found in Appendix A.
- 3 The Unit/Mark setting determines how often the SIR-30 will write a distance mark in the data. Distance marks are used as quick visual indicators of distance. Type in a convenient value.

- 4 Max Units/File:** This is the total expected length of the individual data file. If you know how long you expect the file to be, type in the distance. Otherwise, leave it at 0.
- 5 GPS:** If you are using a GPS, select Custom and enter the location of the GPS receiver with respect to the antenna connected to Channel 1. If it is mounted to the antenna, both the X and Y values will be 0. If the GPS is behind the antenna, the X value will be the distance behind the antenna, and the number will be negative. The Y value determines how far to the right or left the GPS is relative to the antenna connected to Channel 1. If you are standing behind the antenna, facing the direction of your survey profile, this number is positive if the GPS is to the left of the GPR antenna and negative if the GPS is to the right of the GPR antenna.
- 6 Antenna Entries:** Here is where you will enter your antenna selections. The “Antenna 1-4” options correspond to the Channel 1-4 inputs on the back of the SIR-30. For each antenna, choose the Type (Frequency), antenna model (some will only have one available), orientation (angle relative to your profile direction), serial number and antenna location relative to the antenna connected to Channel 1. The X and Y values will be 0,0 for Antenna 1.
- If you are using one antenna, these will be 0,0.
- The serial number is important information if you are using a Model 4105NR 2 GHz Horn with noise reduction technology.
- After you have entered in all of your antennas, click Accept.

Step 2: Collection Settings

For this section, we will refer to the left hand column of the Setup Controls as the Main Menu and the right hand column as the Sub-Menu. The SIR-30 user interface is designed so that you will move through the control settings in a linear manner from top to bottom.

For this basic setup, we will work with the first two buttons on the Main Menu: RADAR CONFIG and SIGNAL PROCESS. When you click one of the Top Menu buttons, it will turn solid blue and the options in the Sub-Menu will change.

If you selected an antenna frequency and model number from the previous screen, the SIR-30 will load many of the important settings as defaults.

- 7** Click RADAR CONFIG > RADAR.
- a)** Ensure that the Run Mode is set to “Distance.”
 - b)** Set Scans/Sec very high. The SIR-30 will default to the maximum value. It is important that this number be at the maximum because it will allow you to collect data faster.
 - c)** Set Samples/Scan to a value high enough to prevent aliasing. A value of 512 or 1024 is usually sufficient. See Appendix B for reasonable values for different combinations of antenna and range.

8 Click RADAR CONFIG > SCAN.

- a) Set the Time Range long enough to reach an appropriate depth for your application. If you are unsure how deep you are reaching, set the range so that the bottom 10-20% of the signal has a “fuzzy, static filled appearance” as in Figure 13. This is noise that results from a heavily attenuated signal and it is useful to have some of it in your data as a marker of maximum penetration.

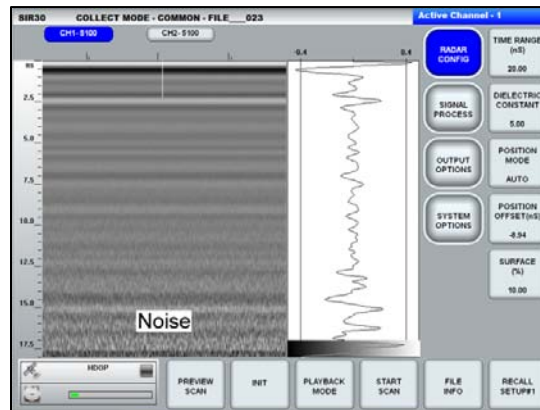


Figure 13: Maximum Time Range (for a 1.6 GHz antenna).

- b) **Dielectric:** The input value has no bearing on depth of penetration. If you know the dielectric constant of the material you are scanning through, you should enter it. This value will also be used for the Migration filter.
- c) **Set the Position and Surface %:** The signal in Figure 13 is properly positioned. Note that the first positive peak of the direct wave is cut by the top of the screen. The direct wave is the first arrival at the receive antenna from the transmit antenna. It represents the beginning of your time range. All GSSI ground coupled, bistatic antennas should be positioned like this. Figure 14 shows two 1.6 GHz ground coupled antennas (left and center) and one 2 GHz Horn antenna (right). The 1.6 GHz antenna on the left is properly positioned with the first strong peak of the direct wave bisected by the top of the window. The center one is positioned too far down the time window. The 2 GHz antenna represents a special case. The direct wave may not be stronger than the ground surface reflection. Be sure to position your Horn antenna so that the direct wave and the ground surface reflection are in the time window. If you need to change it:

- Click on the appropriate channel to activate it.
- Change the Position Mode to Manual.
- Click on Position Offset (nS) and try different values until your scan looks like Figure 13.
- Check Surface % (location of the scan for display purposes) is set to 10.00.

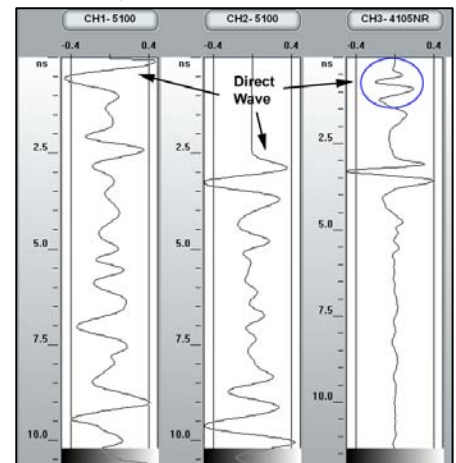


Figure 14: Positioning your Scan.

9 Click RADAR CONFIG > MAINTENANCE.

a) Calibrate Survey Wheel: Click on CALIBRATE SW. If you are using a GSSI supplied survey wheel or DMI, ensure that the SW Mode is Quadrature.

- Mark a measured distance on your survey surface. The longer the distance, the more accurate the calibration. Enter that value in the “Distance” box.
- Position your antenna or vehicle on the Zero of your measured distance and click the Start button.
- Move the antenna/vehicle to the end of you line and click the Stop button. You will see the updated calibration value beneath “New.”
- Repeat the process at least once to make sure you get consistent numbers.

b) Configure your GPS: Make sure the serial cable from your GPS is plugged into the SIR-30 and then click on CONFIGURE GPS. Change the settings in the GPS Serial Port Config menu to match the output from the GPS.

c) Custom Mark: Click CUSTOM MARKS and enter in a Mark Name for any custom data markers that you wish to use.

10 Click SIGNAL PROCESS > GAIN. Be sure to first click the appropriate channel to select it.

- **For Auto Gain:** Make sure that your antenna is on your survey surface and check the o-scope view on the screen. It should look like Figure 15. The scan line should not exceed the screen width nor should it be “flatlined.” If it doesn’t look like Figure 15, click the “INIT” button at the bottom of the screen and it will re-initialize the gains.
- **For Manual Gain:** Set the Gain Mode to “Manual,” select the desired number of points, and add/subtract gain from those points until the signal looks similar to the one in Figure 15.

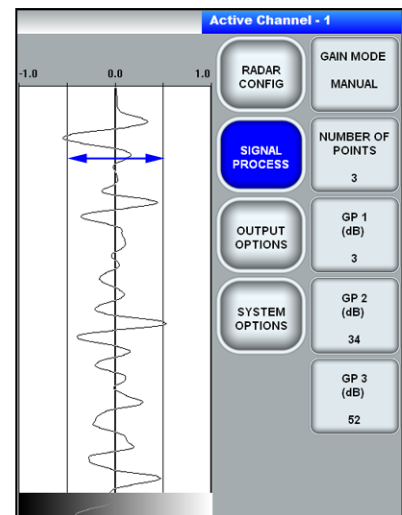


Figure 15: Appropriate Gain.

11 Click SIGNAL PROCESS > FIR FILTERS. The SIR-30 uses FIR filters by default.

- Check filters and change as necessary. If you selected an antenna in the System Setting window, then the SIR-30 loaded default filters. The default Low Pass is 2 or 3 times the center frequency and the default High Pass is $\frac{1}{4}$ to $\frac{1}{3}$ the center frequency. Since the SIR-30 stores raw data and you can apply these filters with different parameters in RADAN, GSSI recommends leaving these at defaults.

12 Collect Data.

- a)** Click on START SCAN and begin collecting. If you are using a survey wheel or DMI it will begin to collect data as soon as you move.
- b)** To collect a new file with the same settings, click on NEXT FILE. The SIR-30 will begin collecting a new file as soon as you click Save/Discard file.
- c)** To stop collecting and return to the setup screen, click STOP FILE. You will be returned to the setup screen after clicking Save/Discard.

Collecting Data with the SIR-30 and Horn Antennas for Road Survey

This section discusses a specific application of the SIR-30: pavement thickness survey with Horn antennas. For this we will assume a setup with only Horn antenna on a survey vehicle. In addition to your GPR equipment, you will need a 4x4 ft (1.2x1.2m) flat metal plate. The plate should be smooth and clean. It can be of any metal or thickness.

Setting Up the SIR-30 for use with a Horn Antenna

This section describes the steps to set up the SIR-30 with the Model 4108 1 GHz, 4105 2 GHz, and the 4105NR 2 GHz Noise Reducing antenna. Be sure that the bottom of your antenna is positioned at least 18" (50cm) from the ground surface.

Note: If you are using a 4105NR, be sure to load the custom filter file onto the SIR-30 before attempting to set up the antenna.

When you turn on the SIR-30 please remember that there is an up to 10 second lag between pushing the "Power On" button on the SIR-30 and a indicator on the SIR-30 that the system is powering up.

Step 1: System Settings

- 1** Specify your desired Scans/Unit and Units/Mark. Type in the expected approximate length of you data files in the Max Units/File box.
- 2** Specify the Type and Model of Horn antenna you are using. If you are using the 4105NR, type in the antenna's serial number in the "Unit S/N" box. You can omit any leading zeroes. This is necessary in order for the SIR-30 to load the custom filter.
- 3** Click Accept.

DISTANCE CONTROL	
Horizontal Units	ft
Scans / Unit	12.00
Units / Mark	5.00
Max Units / File	0.00
GPS	
GPS Type	NONE
ANTENNA 1	
Type	HORN2.0GHz
Model	4105NR
Orientation	0
Unit S/N	155
In-Line(X) Offset	0.00
Cross-Line(Y) Offset	0.00
ANTENNA 2	

Figure 16: Horn Antenna Settings: 4105NR

Step 2: Filters and Gain

- 4** Click SIGNAL PROCESS.
- 5** **Correct Filter Settings:**
 - **For the 4105 NR**, click FIR FILTERS and turn on CUSTOM FILTER. All other IIR and FIR filters must be set to zero.
 - **For the 4105:** FIR High Pass = 250 MHz, FIR Low Pass = 5000 MHz, IIR High Pass = 100 MHz.
 - **For the 4108:** FIR High Pass = 250 MHz, FIR Low Pass = 3000 MHz, IIR High Pass = 100 MHz
- 6** Correct Gain Settings: All Horn antennas should be set to 1 point of Auto Gain.

Step 3: Signal Position

This step is critical. Incorrectly positioning your scan will result in data which cannot be processed. There is no way to fix incorrectly positioned data after it has been collected. Any errors made in this step will mean that you will have to re-collect the data. The data must be recorded with the direct wave and the ground surface reflection.

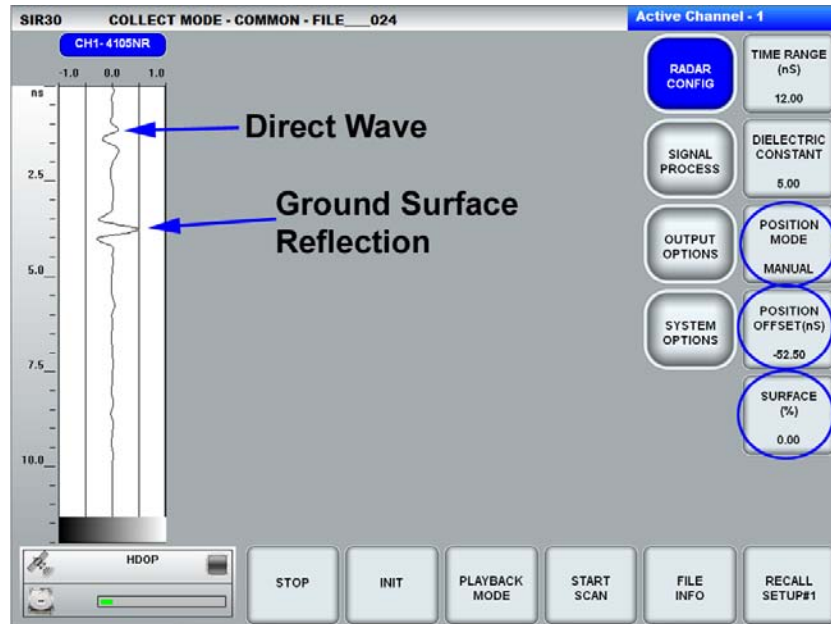


Figure 17: Horn signal position

- 7** Click RADAR CONFIG > SCAN.
- 8** Set the POSITION MODE to Manual.
- 9** Set the SURFACE (%) to 0.00.
- 10** Change POSITION OFFSET until the scan trace looks like Figure 17. It is very important that there be a short amount of “flatline” data above the Direct Wave.

Step 4: Calibrate the DMI or Survey Wheel

- 11** Click RADAR CONFIG > MAINTENANCE.
- 12** Calibrate Survey Wheel. Click on CALIBRATE SW. If you are using a GSSI supplied survey wheel or DMI, ensure that the SW Mode is Quadrature.
- 13** Mark a measured distance on your survey surface. The longer the distance, the more accurate the calibration. Enter that value in the “Distance” box.
- 14** Position your antenna or vehicle on the Zero of your measured distance and click the Start button.
- 15** Move the antenna/vehicle to the end of your line and click the Stop button. You will see the updated calibration value beneath “New.”
- 16** Repeat the process at least once to make sure you get consistent numbers.

Step 5: Define Custom Marks (Optional)

- 17** Click RADAR CONFIG > POSITIONING > CUSTOM MARKS.
- 18** Enter a Mark Name beside your desired Mark Key.

Step 6: Configure GPS (Optional)

- 19** Please see the section of Configuring the SIR-30 to work with GPS.

Collecting Data

This section describes what happens immediately following software setup.

Step 1: The Metal Plate Calibration File

For road thickness measurement, we need to collect a special type of file first. This file will be collected with the antenna moving up and down while positioned over a metal plate. This will allow the SIR-30 to record all of the transmitted energy at different antenna elevations. This file will be used by the RADAN software to generate accurate GPR transmission velocities in the pavement (to give accurate thicknesses) and to remove any confusion in the data related to antenna bounce.

- 1** Center the metal plate under the antenna. You will notice the ground surface reflection get very strong.
- 2** Click RADAR CONFIG > RADAR. Set the RUN MODE to TIME. We need to collect this first file without the survey wheel/DMI.
- 3** Click the INIT button to reset the gains. The ground surface reflection will weaken.
- 4** Have one person begin to jump up and down on the bumper where the antenna is located.
- 5** Once the antenna is moving up and down, click START SCAN and collect approximately 10 seconds of data.
- 6** After 10 seconds, click STOP SCAN and save the data.
- 7** Remove the metal plate.
- 8** Click RADAR CONFIG then RADAR. Set the RUN MODE to back to DISTANCE.
- 9** Click the INIT button reset the gains. The ground surface reflection will strengthen.
- 10** Be sure to note the file name of the metal plate calibration file.

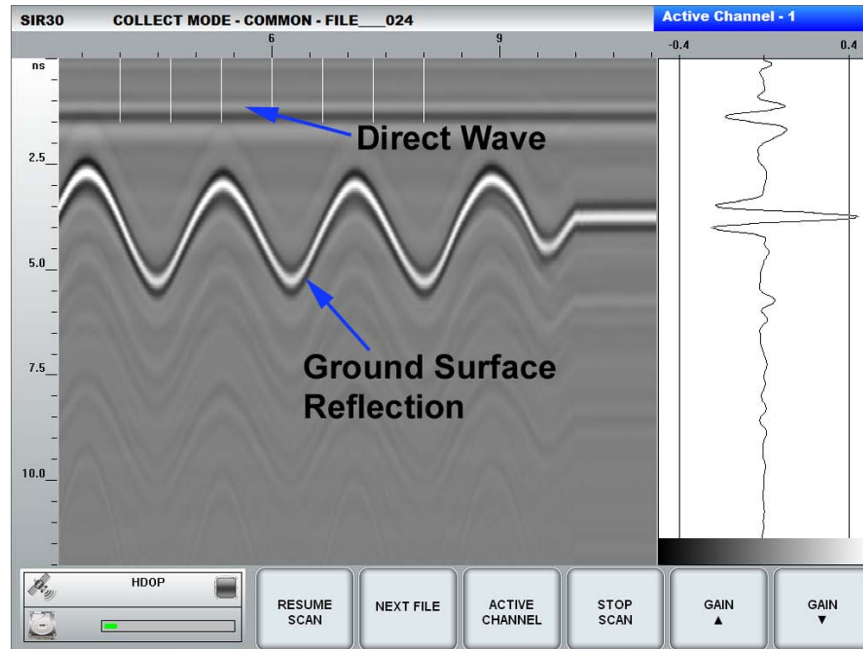


Figure 18: Calibration File Collection

Step 2: Change Units (Optional)

- 11** You may prefer to collect with your horizontal units set to miles (km).
- a)** Click OUTPUT OPTIONS > SYSTEM UNITS.
 - b)** Select DISTANCE UNITS and choose your desired unit.

Step 3: Change Color Table (Optional)

- 12** Pavement data may be easier to see if a high contrast gray scale color table.
- a)** Click OUTPUT OPTIONS > DISPLAY.
 - b)** Click COLOR TABLE and select Color Table 3. This is the bar in the center.
 - c)** Click COLOR XFORM and select Color Xform 1, the first bar on the list.

Step 4: Starting your Profile from a Non-Zero Point (Optional)

- 13** You may wish to start a data file from a specific mile (km) mark and either count up or down. This must be done for each new file you create. For example, if you start it at 10, counting up and go one mile, the next file will not automatically begin at 11. You will need to re-enter a starting position.
- a)** Click FILE INFO in the control bar at the bottom of the screen. You will see the screen in Figure 19.
 - b)** Enter your starting point in the START POS box.
 - c)** If your mile marks increase, select COUNT UP in the DIRECTION chooser.

SIR30 COLLECT MODE - COMMON - FILE 024 Active Channel - 1

COLLECT FILE INFO

FILE NAME: FILE_024

START POS (mi): 7

DIRECTION: COUNT UP

COMMENTS: Route66

RADAR PARAMETERS

Created: 3/10/2011 10:34 AM
Transmit Rate: 200 KHz
Samples/Scan: 512
Scan Rate: 90
Dielectric: 5

CH1- Antenna: 4105NR
Time Range: 12 nS
Position Correction: -52 nS

PROCESSING HISTORY

CH1- Range Gain (dB): 16
Custom Filter

ACCEPT CANCEL

Figure 19: Start Pos/Direction Inputs.

Step 5: Collect Data

- 14** Click on START SCAN and begin collecting. If you are using a survey wheel or DMI it will begin to collect data as soon as you move.
- a)** To collect a new file with the same settings, click on NEXT FILE. The SIR-30 will begin collecting a new file as soon as you click Save/Discard file.
 - b)** To stop collecting and return to the setup screen, click STOP FILE. You will be returned to the setup screen after clicking Save/Discard.

Chapter 4: Data Playback

The SIR-30 is designed to collect an extremely large quantity of raw data at high speed. The data you see on the screen during collection has been image processed to show the effect of the various filters and gain you have applied. Data playback is somewhat limited due to the way these are applied. As a result, you will only be able to scroll forward and pause data playback, but not scroll backwards. If you are running the SIR-30 from a separate laptop with RADAN installed, you may find it more convenient to view the data in RADAN rather than attempt to use the SIR-30 playback mode.

Step 1: Enter Playback Mode

- 1 Click on the PLAYBACK MODE button on the Control Bar.

Step 2: Choose File

- 2 Click on the file that you would like to view and then click SELECT. You will see a check show up in the box to the left of the file name. Click ACCEPT. You will see “Playback Mode” at the top of the screen and the Control Bar will now look like Figure 20.



Figure 20: Playback Setup Control Bar

Step 3: Playback Data

- 3 Click the PLAYBACK FILES(S) button to display your data. You will see the Control Bar change to look like Figure 21. The data will scroll across the screen. Click PAUSE to stop the scrolling. GAIN (UP) will increase screen contrast and may make weaker reflections easier to see. If you collected data with more than one channel, click ACTIVE CHANNEL to display the other channels.



Figure 21: Playback Run Control Bar

- 4 Click STOP PLAYBACK > COLLECT MODE to return to collect mode and record new data.

Chapter 5: Data Transfer

All data are stored on the SIR-30 internal memory, even if you are using a separate laptop to control it.

Transferring Data from the SIR-30 to the Control Laptop

This section assumes that you are using a separate laptop to control the SIR-30. Data will first be copied to the laptop and can be then copied off of the laptop or processed on the laptop if you have a copy of RADAN7 installed.

Step 1: Create a Folder on your Laptop

- 1** Exit the SIR-30 control program or push the Windows button on the laptop keyboard and click Computer from the popup menu.
- 2** Navigate to the Local Disk (C:).
- 3** Right-click in the Explorer window and select New > Folder.
- 4** Name the new folder “radardata”. If there is already one on the “C” drive, you can delete the new one you just made.
- 5** Re-open the SIR-30 program.

Step 2: Move Data

- 6** Click OUTPUT OPTIONS > MANAGE FILES.
- 7** Make sure you are in the Data Path where your desired files are located. If not, click DATA PATH and choose it from the list.
- 8** Click COPY TO LAPTOP. GSSI recommends “copy” as opposed to “move” because “move” will delete the data from the SIR-30. It is best to copy the data, confirm that everything is OK, and then delete the data from the SIR-30 later.
- 9** Choose the files you wish to copy and click accept. If you have collected multi-channel data files, you can click SPLIT CHANNELS and they will be broken up into separate data files.
- 10** You can now transfer your data to another computer via a removable hard drive, CD/DVD or USB flash drive.

Step 3: Confirm and Delete

- 11** Return to the folder on the “C” drive and confirm that your data files have transferred.
- 12** Return to the SIR-30 program and select OUTPUT OPTIONS > MANAGE FILES > DELETE FILES.
- 13** Select the files you wish to delete and then click Accept. The SIR-30 will not ask for confirmation, so be sure those files are the ones you want to delete.

Transferring Data from the SIR-30 to an External Drive

This section assumes that you have been running the SIR-30 directly and that you wish to copy the data to a removable drive or USB flash.

A note on the SIR-30 drive structure: When you open the OUTPUT OPTIONS submenu, you will see that there is an option for Drive E and Drive F. These are not mapped to specific USB or Ethernet drive. The first device you plug into the SIR-30 will be classified as Drive E. Advanced Windows users will be able to map a particular drive letter to a specific device.

Step 1: Create a Folder on your Laptop.

- 1** Push the Windows button on the keyboard and click Computer from the popup menu.
- 2** Navigate to the Removable Drive/USB Disk (E:).
- 3** Right click in the Explorer window and select New > Folder.
- 4** Name the new folder “radardata”. If there is already one on the “E” drive, you can delete the new one you just made.
- 5** Re-open the SIR-30 program.

Step 2: Move Data

- 6** Click OUTPUT OPTIONS > MANAGE FILES.
- 7** Make sure you are in the Data Path where your desired files are located. If not, click DATA PATH and choose it from the list.
- 8** Click COPY TO DRIVE E. GSSI recommends “copy” as opposed to “move” because “move” will delete the data from the SIR-30. It is best to copy the data, confirm that everything is OK, and then delete the data from the SIR-30 later.
- 9** Choose the files you wish to copy and click accept. If you have collected multi-channel data files, you can click SPLIT CHANNELS and they will be broken up into separate data files.
- 10** You can now move the drive to another computer.

Step 3: Confirm and Delete

- 11** Return to the folder on the “C” drive and confirm that your data files have transferred.
- 12** Return to the SIR-30 program and select OUTPUT OPTIONS > MANAGE FILES > DELETE FILES.
- 13** Select the files you wish to delete and then click Accept. The SIR-30 will not ask for confirmation, so be sure those files are the ones you want to delete.

NOTE: If you get an error window saying “The system cannot find the path specified” it is because you have misnamed or forgotten to create a folder called “radardata” on your destination drive/device.

Chapter 6: Configuring the SIR-30 to Work with a GPS

The SIR-30 is designed to work with a wide variety of GPS units. In order to operate with the SIR-30, your GPS must be able to output the NMEA GGA string through a 9-pin serial port.

The GPS data are saved in a separate file with the same root name as your GPR data file. A profile collected with GPS will thus have three files associated with it: FILE____001.DZT (RADAR data), FILE____001.DZW (filter and gain settings for the raw data), and FILE____001.DZG (GPS location info).

Step 1: Hardware Setup

- 1** Securely attach your GPS receiver to your survey vehicle or antenna.
- 2** Measure the in-line (front-back) and cross-line (left-right) offset from the center of the antenna connected to channel 1 to your GPS receiver.
- 3** Plug the serial port from your GPS into the GPS port on the back of the SIR-30.
- 4** The yellow LED on the front of the SIR-30 will flash if it detects an incoming GPS message.

Step 2: Initial System Settings

- 5** You will see this screen (Figure 22) on startup. You can also access it by clicking RADAR CONFIG > SYSTEM SETTINGS.

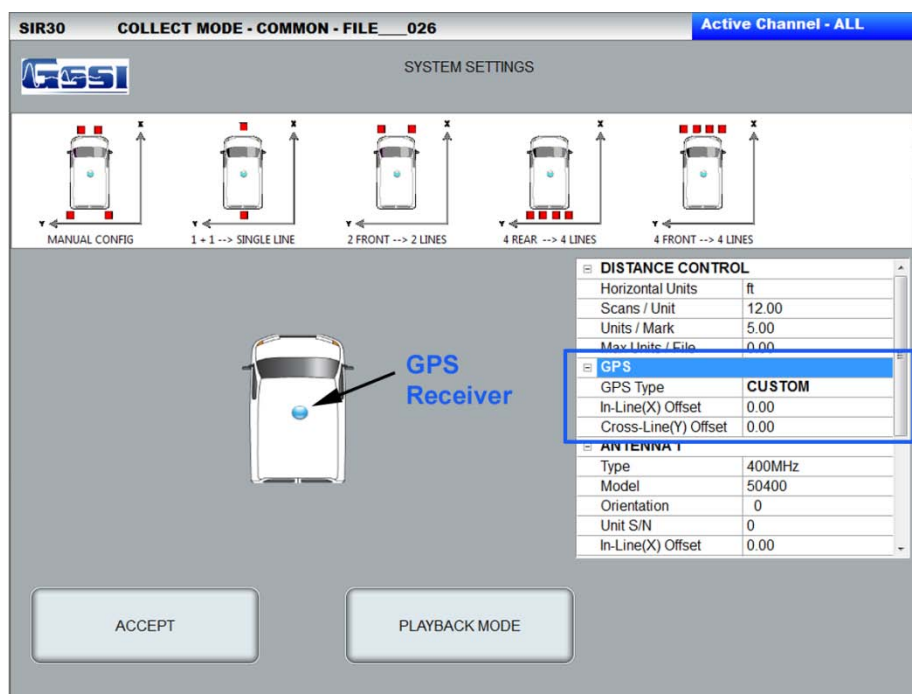


Figure 22: GPS Initial Hardware Setup.

- 6** Change GPS Type to “CUSTOM.”
- 7** Enter the In-Line (front-back) and Cross-Line (left-right) distance from the center of the antenna connected to channel 1. If the antenna is in front of the GPS receiver, the In-Line value is negative. If it is to the left of the antenna (facing the direction of survey) than the Cross-Line value is positive.
- 8** Click Accept.

Step 3: Matching the GPS Output Settings

- 9** Select RADAR CONFIG > POSITIONING > CONFIGURE GPS. You will see the screen shown in Figure 23.

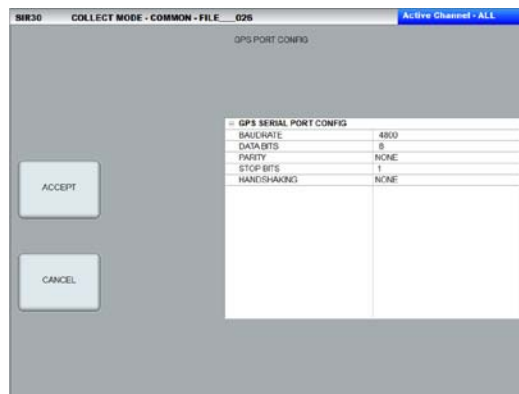


Figure 23: GPS Port Config.

- 10** The five options shown in Figure 23 have pull-down menus. Change these settings to match the settings on your GPS. GSSI most likely does not have experience with your particular type of GPS, so it is up to you to locate and understand these settings within your GPS device.
- 11** Click Accept.

Step 4: Check GPS Data Integrity

- 12** The SIR-30 performs a constant quality check on the incoming GPS stream. This is visible at the lower left-hand corner during collection and setup. The SIR-30 checks the HDOP once per second. If the HDOP is less than 3, the quality control square will be green, indicating a high quality signal. If the HDOP is 3 or greater, it will be yellow indicating a low quality signal. If the SIR-30 loses communication with the GPS, it will turn red. If the SIR-30 doesn't receive a data feed from the GPS for more than 10 seconds, including stopping at intersections, it will alarm with a tone.

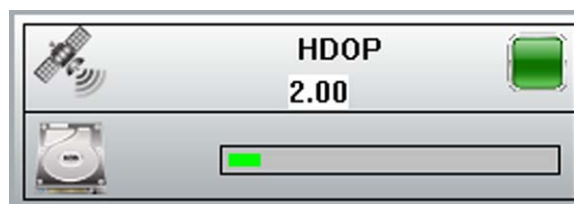


Figure 24: GPS Quality Check.

Appendix A: Common Survey Parameters

This appendix lists some recommended parameters for different applications. These settings represent a good starting point and you should feel free to modify them as needed.

Pavement Thickness Mapping

Antenna: Models 42000 or 4105NR 2 GHz Horn antenna

Range: 12-15 nS

Gain: 1 pt Auto

Scans/Unit: 1/ft for High Resolution. 5Ft/Scan for Low Resolution.

Bridge Deck Deterioration Mapping

Antenna: Model 5100B 1.6 GHz

Range: 12-15 nS

Gain: 1 pt Auto

Scans/Unit: 24/ft

Underground Utility Locating/Archaeological Survey

Antenna: Model 50400 400 MHz, Model 5104 270 MHz.

Range: 30-150 ns

Gain: 3-5 pt Auto

Scans/Unit: 24/ft for High Resolution. 12/ft for Low Resolution.

Geotechnical Investigation

Antenna: Model 5106 200 MHz, Model 3207 100 MHz

Range: 150 -500 nS

Gain: 3-8 pt Auto

Scans/Unit: 6/ft for High Resolution. 1/ft for Low Resolution.

Concrete Investigation

Antenna: Model 5100B 1.6 GHz, Model 62000 2.0 GHz Palm, Model 52600 2.6 GHz

Range: 12-15 nS

Gain: 2 pt Auto

Scans/Unit: 120/ft for High Resolution. 60/ft for Low Resolution.

Appendix B: Antenna Specifications

Model #	Center Frequency (MHz)	Pulse Width (nS)	Max PRF* (*KHz)
4105(NR)/42000	2000	0.5	300
50400A	400	2.5	500
50400	400	2.5	250
5104	270	3.7	200
5106	200	5	200
5100B	1600	0.6	200
52600	2600	0.4	200
4108	1000	1	300
3101A	900	1.1	100
3200MLF	16-80		75
3207	100	10	50

Appendix C: Dielectric Values of Common Materials

Material	Dielectric	Velocity (mm/ns)
Air	1	300
Water (fresh)	81	33
Water (sea)	81	33
Polar snow	1.4 – 3	194 - 252
Polar ice	3 - 3.15	168
Temperate ice	3.2	167
Pure ice	3.2	167
Freshwater lake ice	4	150
Sea ice	2.5 – 8	78 – 157
Permafrost	1 – 8	106 – 300
Coastal sand (dry)	10	95
Sand (dry)	3 – 6	120 - 170
Sand (wet)	25 – 30	55 – 60
Silt (wet)	10	95
Clay (wet)	8 – 15	86 – 110
Clay soil (dry)	3	173
Marsh	12	86
Agricultural land	15	77
Pastoral land	13	83
“Average soil”	16	75
Granite	5 – 8	106 – 120
Limestone	7 – 9	100 – 113
Dolomite	6.8 – 8	106 – 115
Basalt (wet)	8	106

Shale (wet)	7	113
Sandstone (wet)	6	112
Coal	4 – 5	134 – 150
Quartz	4.3	145
Concrete	5 – 8	55 – 120
Asphalt	3 – 5	134 – 173
PVC	3	173

**Meter to English conversion factor: 2.54 cm in 1 inch.

** Table of Dielectric values adapted from:

Reynolds, John M.
1997 *An Introduction to Applied and Environmental Geophysics*, John Wiley & Sones, New York.

Appendix D: SIR-30 Maintenance Tool Overview

You have been provided a SIR-30 Maintenance Tool on a GSSI USB flash drive marked “S30 MAINT”. This utility is to be kept in a safe location. Although you will be able to download the utility from the website, we wanted to provide it directly so you wouldn’t need to go looking for it. It provides many troubleshooting tools for both the SIR-30 host as well as client computers including:

SIR-30 Update Finder: If you plug the utility into a web enabled computer it will find the latest SIR-30 UI and firmware versions on our website and ask you if you want to download it to the maintenance utility. Each version released will come with a document explaining what steps need to be taken to push the changes to the SIR-30.

SIR-30 Updater: If you plug the utility into the SIR-30 and press the “Update SIR-30 UI” it will allow you to update the software on the SIR-30 unit.

SIR-30 Firmware Updates: Pressing the “Prep System for Firmware Updates” button will copy the firmware from the maintenance utility to the SIR-30 and open the SIR-30 UI in an update mode. It is only from this utility that you will be able to open the versions screen and make changes to your firmware.

SIR-30 Client Setup: The maintenance utility can be used to set up any Windows 7 or Vista computer to act as a SIR-30 client. (A separate document will be provided to explain this functionality).

SIR-30 Troubleshooting: Nearly any button can be pressed within the utility to help issues you may have run into. Any change that could have an adverse effect is explained before you are allowed to continue (at your option).

To run it: Just insert it into your SIR-30 (or other computer), Open it, Right click on “SIR-30_MaintenanceTools” and select “Run as administrator”.

What it Isn’t

Not a Recovery Disk: This tool is not a recovery disk for the operating system of the SIR-30. If there is a hardware issue that results in a disk failure in the field the SIR-30 would need to be returned to an approved GSSI service center (most likely GSSI itself).

Not a Data Transfer Utility: This tool is not intended for data transfer of dzt files.

Not a USB Security Key: This tool is not a USB security key like the keys previously shipped with RADAN 6.6 or earlier.

Obtaining the Latest SIR-30 Maintenance Utilities Distribution

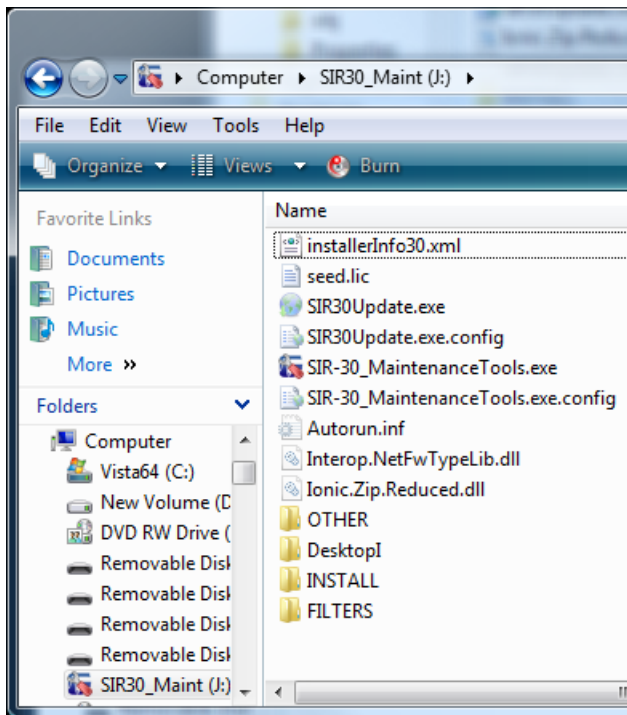
You may have received the maintenance tool with your SIR-30 unit. You can also download the latest SIR-30 Maintenance package from support.geophysical.com. If you need to download it see the instructions below:

Requirements:

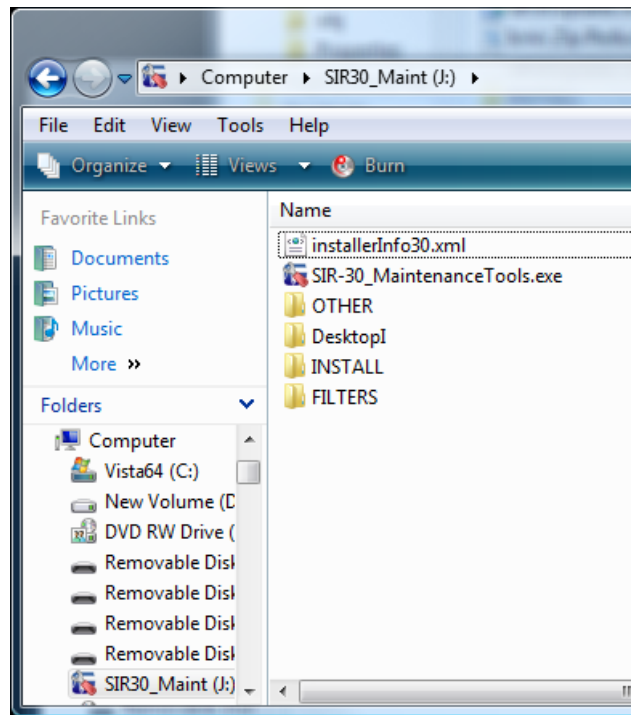
- Any Windows Vista or Windows 7 computer **connected to the internet**
- A Clean USB Flash Drive (512Mb or greater in size recommended).

Instructions:

- 1 Open a browser and go to <https://support.geophysical.com/gssiSupport/SIR30Maint.zip>
- 2 Save the file to your computer when prompted.
- 3 Unzip/extract package to a clean USB flash drive. Upon completion, you should have a directory that looks like one of the following (it depends on your Explorer folder options set in the Operating System – For instance: The SIR-30 will display the Right version):



Left: Hidden files are visible in Folder Options



Right: Hidden files stay hidden in Folder Options

Obtaining the Latest SIR-30 Software/Firmware Distribution

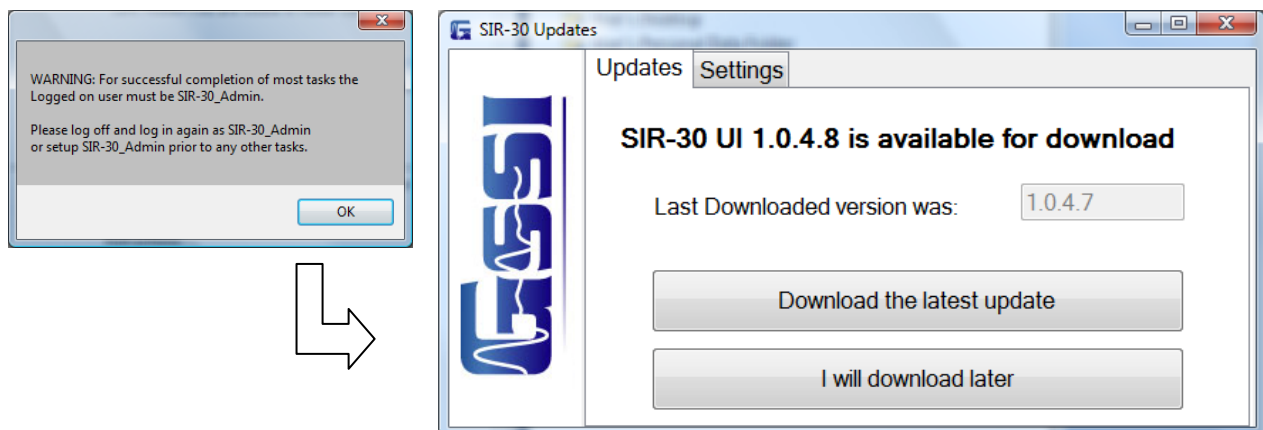
The maintenance tool contains a built in handy utility that will go out and search the support site for the latest version of SIR-30 software and firmware.

Requirements:

- Any Windows Vista or Windows 7 computer **connected to the internet**
- A user with **admin privileges**.

Instructions:

- 1** Insert the SIR-30_Maint tool into a computer that can connect to the internet.
- 2** Open the SIR30_Maint drive and
- 3** Right click SIR-30_MaintenanceTools.exe and run as admin.
- 4** The software will automatically attempt to locate the latest version of SIR-30 software and firmware and compare it to the distribution currently on the maintenance tool.



Note: It may take 30 seconds after you press OK to bring up the SIR-30 Updates Dialog. If a newer version is found it will ask you if you want to download the latest version.

- 5** Press “Download the latest update”. **Note:** This downloads the update files to your USB flash drive and unpacks them in the appropriate directories. It does NOT install them to your SIR-30 at this time. That feature is covered under a separate section of this document.

Note: After it finishes you may notice that the 1.0.4.8.zip (or similar) is located on your flash drive. As it should have unpacked properly you should be able to delete it by hand if you desire. Or you can leave it until you decide to do a future clean up. Some users may prefer to keep the past few zips as archives in case they want to restore a previous version.

Appendix E: SIR-30 Wiring Instructions

The SIR30 runs with a DC input from +10VDC to +28VDC. It will not turn on unless there is a minimum of +10V at the power connector. The system will shut down immediately if the input drops below +9V under load. The input DC current requirement is a function of the number of antennas and the input voltage. For reliable operation the wiring should support 300W to the unit, although typical power consumption will be much less. This is 30A at the minimum voltage.

The AC Supply provided with the unit produces +24V when connected to a wide range of mains voltages.

The DC input connector has 5 pins assigned as follows:

PIN A = INPUT +

PIN B = INPUT +

PIN C = INPUT – (GND)

PIN D = INPUT – (GND)

PIN E = IGNITION

In order for the SIR30 to turn ON the INPUT pins should all be wired as indicated and the IGNITION pin needs to be connected to a voltage in the range of +5V to +INPUT.

Wiring Options:

AC Power:

In this mode the AC power supply is connected to the SIR30 Power connector. Mains voltage should be supplied to the AC supply at all times while the unit is in use. The Power Switch on the SIR30 Front Panel should be used to turn the system on and off.

Do not interrupt power to the AC supply while the system is in operation.

DC Battery Cable – no ignition wiring

The DC Battery Cable has two clips that can be used to power the system. Be sure to connect the RED clamp to the POSITIVE (+) terminal of the battery. The BLACK clamp connects to the NEGATIVE (-) terminal of the battery. The Power Switch on the SIR30 Front Panel should be used to turn the system on and off.

The power connector back shell has a short wire loop connecting a green wire plug (P1) with a red wire jack (J2). Leave this loop connected. Do not remove either clamp while the system is in operation.

DC Battery Cable – direct wiring with ignition connection

The DC Battery Cable has two nuts that attach the battery clips to the cable. If those nuts are removed, the cable can be attached directly to the battery lugs or a terminal block that is supplied with power all of the time the unit is in use.

With this wiring option it is recommended that the IGNITION terminal be connected. This will prevent the unit from draining the vehicle battery if inadvertently left on. If the SIR30 Power Switch is left in the ON position, the SIR30 will turn on and off with the vehicle ignition.

To connect the ignition line, disconnect J2 and P1. Connect the P1 (green wire) to a DC source that turns on and off with the vehicle. Leave J2 disconnected.

Wiring Requirements:

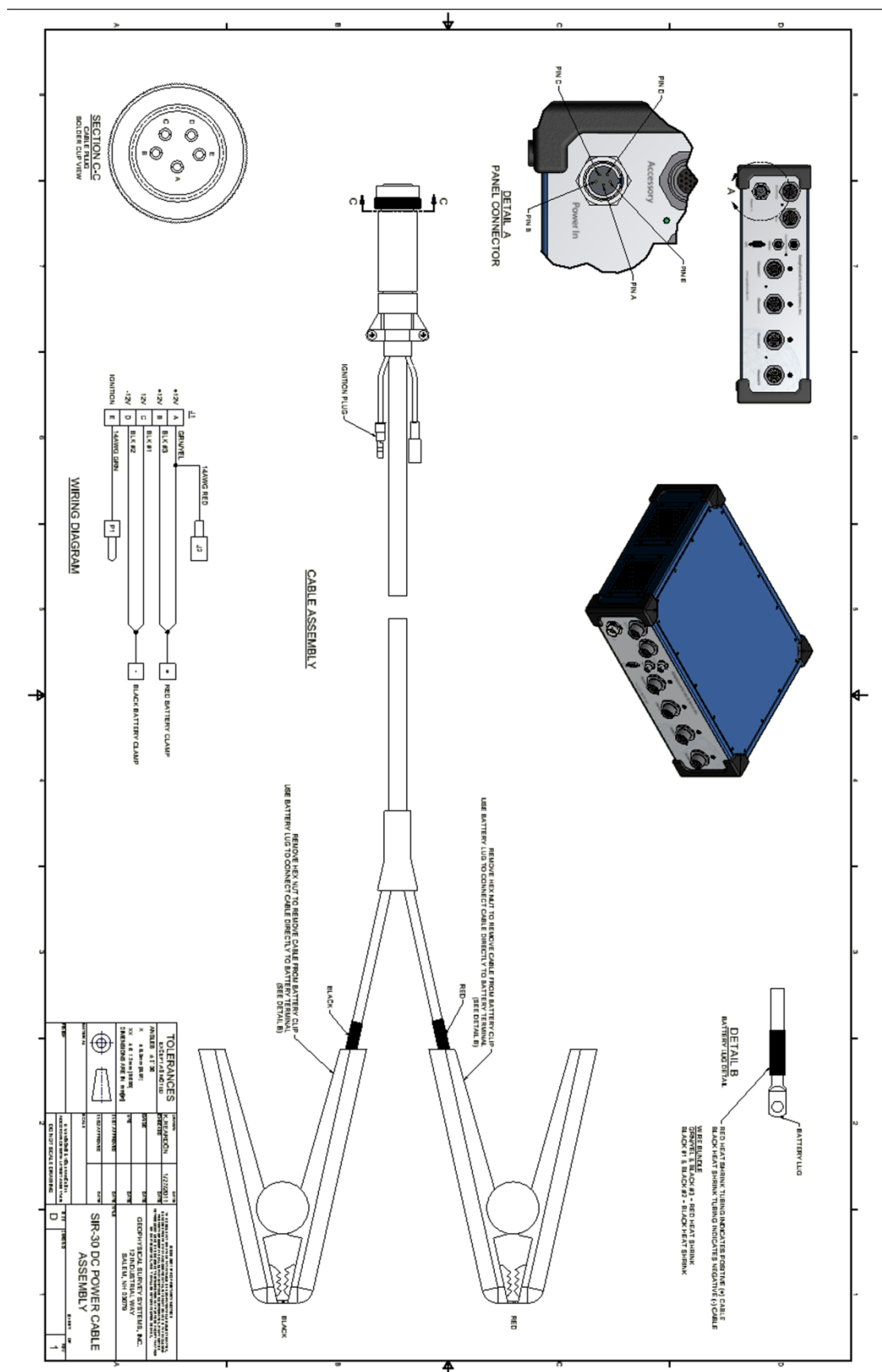
If custom wiring is to be used to power the unit in a +12V application the wire must be rated for 30A and all 4 INPUT terminals wired. The GSSI cable uses 4 x 12AWG stranded conductors and this should be suitable for 20 feet of cable length.

General Notes about Vehicle Wiring:

It is quite common for several pieces of vehicle mounted equipment to be interconnected by signal cable in addition to power connections. In this instance it is strongly recommended that all equipment use a common ground point. This will minimize the chances of improper operation or damage occurring due to voltage drops around the vehicle.

Good practice is to use a fused connection box to supply power to several pieces of equipment. In this case a large cable is used to bring power to the box and all equipment is powered from the box. This results in all equipment having a common ground point. Be sure to use a cable capable of handling the current required by all of the connected equipment.

Never connect/disconnect equipment that is running. This can create sparks that can damage all connected equipment.



Appendix F: Glossary of Terms and Suggestions for Further Reading

Antenna: a paired transmitter and receiver that sends electromagnetic energy into a material and receives any reflections of that energy from materials in the ground. Also called a transducer. Antennae are commonly referred to by their center frequency value (i.e. 400MHz, 1.5Ghz). This frequency determines the depth of penetration and the size of the objects or layers visible.

Attenuation: the weakening of a radar pulse as it travels through different materials.

Center Frequency: the median transmit frequency of an antenna. The antenna will also transmit energy at a frequency range of 0.5-2 times its center value. For example, a 400 MHz antenna may actually transmit at a range from 200-800 MHz.

Clipping: occurs when the amplitude of a reflection is greater than the maximum recordable value. The system disregards the true value of the reflection and writes in the maximum allowable value. Clipping appears in the O-Scope as signal that “goes off the scale” at the sides of the window.

Dielectric permittivity: the capacity of a material to hold and pass an electromagnetic charge. Varies with a material’s composition, moisture, physical properties, porosity, and temperature. Used to calculate depth in GPR work.

EM: Acronym for electro-magnetic.

FCC: Acronym for Federal Communications Commission. The United States governmental body that oversees the UWB industry of which GPR is a part.

Gain: amplifying the signal to certain section of a radar pulse in order to counteract the effects of attenuation and make features more visible.

GHz: Acronym for Gigahertz. A measurement of frequency equal to one billion cycles per second.

GPR: Acronym for Ground Penetrating Radar.

Ground-coupling: the initial entry of a radar pulse into the ground.

Hyperbola: an inverted “U.” The image produced in a vertical linescan profile as the antenna is moved over a discrete target. The top of the target is at the peak of the first positive (white in a grayscale color table) wavelet.

Interface: the surface separating materials with differing dielectric constants or conductivity values.

KHz: Acronym for Kilohertz. A measurement of frequency equal to one thousand cycles per second.

Linescan: commonly used method of depicting a radar profile. Linescans are produced by placing adjacent scans next to each other and assigning a color scheme to their amplitude values.

Macro: a preset list of processing options that may be applied to perform repetitive functions on an entire dataset. Macros may be created and edited to include different functions (see RADAN manual for addition information).

Mark: point inserted along a survey line manually by the operator or at preset intervals.

MHz: Acronym for Megahertz. A measurement of frequency equal to one million cycles per second.

Migration: mathematical calculation used to remove outlying tails of a hyperbola and to accurately fix the position of a target.

Nano-second: unit of measurement for recording the time delay between transmission of a radar pulse and reception of that pulse's reflections. Equal to one one-billionth of a second.

Noise: unwanted background interference that can obscure true data.

Noise floor: time depth at which the noise makes target identification impossible.

nS: see Nano-Second.

Oscilloscope: device used to view and measure the strength and shape of energy waves. Common term in GPR industry for a method of data display showing actual radar wave anatomy.

Passband: the frequency range at which the antenna is emitting energy. It is roughly equivalent to 0.5-2 times the center frequency.

RAM: Acronym for random access memory. Temporary memory in which a computer stores information used with a running program, or temporarily stored data before it is written to a hard drive.

Range: the total length of time (in nanoseconds) for which the control unit will record reflections. Note: indicates two-way travel time.

RF: Acronym for radio frequency.

Sample: a radar data point with two attributes: time and reflection amplitude. A third attribute, position, is assigned by the user. Under-sampling will produce a scan wave that does not contain enough information to draw a smooth curve. It may miss features. Over-sampling will produce a larger data file.

Samples/Scan: the number of samples recorded from an individual radar scan. Commonly set to 512.

Scan: one complete reflected wave from transmission to reception, sometimes called a trace.

Survey wheel: wheel attached to an antenna and calibrated to record precise distances. Necessary for accurate data collection.

Time-slice: a horizontal planview of amplitude values drawn from adjacent vertical profiles. The time-slice is produced for a particular time-depth and is vital for understanding the horizontal positions of features in a survey area.

Time window: the amount of time, in nano-seconds, that the control unit will count reflections from a particular pulse. Set by the operator.

Transect: a line of survey data. An area is systematically surveyed by recording transects of data at a constant interval. The transects are then placed in their correct position relative to each other in a computer and horizontal time-slices are produced.

UWB: Acronym for Ultra-Wide Band. Refers to the wide frequency band of emissions put out by a GPR device.

Wiggle trace: method of GPR data display showing oscilloscope trace scans placed next to each other to form a profile view. Commonly used method in seismic studies.

Further Reading

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